

TIAN YU

buc9hh@virginia.edu $\diamond$ 776styjsu.github.io

Research Interests

My research develops testing and runtime monitoring methods for AI-enabled autonomous systems, with a focus on using model reasoning and behavior to identify failures and assess safety.

Education

University of Virginia

Ph.D. Student in Computer Science
Advisor: Sebastian Elbaum

Aug 2025 – Present
Charlottesville, VA

University of Washington

B.S. in Computer Science; GPA: 3.7/4.0

Sept 2021 – Jun 2025
Seattle, WA

Publications and Preprints

Tian Yu, Lu Feng, and Sebastian Elbaum. “Drive the Thoughts: Runtime Monitoring of VLA Reasoning-Trajectory Consistency.” 2026. *Manuscript under review*. [\[Paper\]](#) [\[Code\]](#)

Research Experience

Graduate Research Assistant, University of Virginia

Aug 2025 – Present

Runtime Monitoring of Reasoning–Trajectory Consistency

Jan 2026 – Present

- Investigated whether driving VLA (Alpamayo 1.5) reasoning can serve as a behavioral specification for runtime monitoring.
- Created a benchmark of **150 manually annotated reasoning–trajectory pairs** assessing reasoning reliability, reasoning–trajectory consistency, and trajectory safety.
- Developed runtime monitors that incorporate road geometry through lane-relative trajectory representations, achieving $F_1 = 0.75$ for reasoning–trajectory inconsistency detection (**+0.13** over the strongest baseline).

Reasoning Coverage for Testing Autonomous Driving Models

Aug 2026 – Present

- Developing a coverage framework that represents VLA reasoning as structured paths linking scene observations, causal relations, and driving decisions.
- Evaluating whether reasoning-based test prioritization exposes distinct safety failures beyond those found using scenario- and trajectory-based coverage.

Attribution Stability and Safety in Autonomous Driving

Aug 2025 – Present

- Investigated whether unstable visual attributions reveal driving risks overlooked by stable steering predictions.
- Developed adversarial perturbations to amplify saliency changes while limiting image distortion and steering deviations.
- Evaluated attribution instability in closed-loop **CARLA** simulations, finding limited predictive value for near-term failures.

Undergraduate Research Assistant, University of Washington

Advisor: Michael D. Ernst

Jan 2023 – Jun 2025

Improving Unit Test Generation via Program Analysis [\[Code\]](#)

Jan 2023 – Jun 2025

- Extended **Randoop** with type-dependency analysis to construct missing test inputs and side-effect analysis to generate diverse object states.
- Built an automated evaluation pipeline across 30+ Java projects to measure generated tests' code coverage and fault-detection effectiveness using mutation testing.

Cross-Language API Exploration via Static Analysis [\[Report\]](#) [\[Code\]](#)

Jan 2025 – Mar 2025

- Built **Jungle-Mapper**, a static analysis tool that extracts type and API relationships from source code to discover API call sequences for constructing desired object types.
- Designed a shared graph representation and query engine for Java and Rust, evaluating tradeoffs between cross-language extensibility, development effort, and solution correctness.

Industry Experience

Software Engineering Intern

China Pacific Insurance Company

Topic: LLM Application Development

Jul 2023 – Sept 2023

Shanghai, China

Honors and Awards

Provost's Fellowship, University of Virginia

Dean's List, University of Washington (multiple quarters)

Technical Skills

Programming Languages: Python, Java, Bash

Development & Infrastructure: Git, Linux, Docker

Simulation Platforms: AlpaSim, CARLA